

Your LAST Name _____

Your FIRST Name _____

Your Collaborators

• Name: _____ SID: _____

• Name: _____ SID: _____

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Policy Statement

- We encourage you to collaborate, but only in a group of up to *five* current EE 120 students.
- Be sure to write the names of your collaborators at the appropriate location on the next page. Each collaborator must submit a distinct, self-prepared solution document containing original contributions to the collaborative effort.
- To submit your work, you have several options:
 - (a) Print out this homework document and write your solutions on the printout. Then upload a *PDF file* of the scan.
 - (b) Write your solutions on this PDF file using a tablet device. Then upload a *PDF file*.
 - (c) Use your own sheets of paper, but demarcate clearly the space for each problem, part, and subpart. Then upload a *PDF file*.
- Please write neatly and legibly, because *if we can't read it, we can't evaluate it*. Remember that, even though the problem sets are self-graded, we will evaluate your work whenever we check your self grades.
- Unless we explicitly state otherwise, you may expect full credit *only if* you explain your work succinctly, but clearly and convincingly.
- If you are asked to provide a “sketch,” it refers to a well-labeled *hand-drawn* sketch—not a plot generated by a computing device.
- On occasion, a problem set contains one or more problems designated as “optional.” Your solutions to such problems will NOT be evaluated. Nevertheless, you’re responsible for learning the subject matter within their scope.
- If you have a confirmed disability that precludes you from complying fully with these instructions or with any other parameter associated with this problem set, please alert us immediately about reasonable accommodations afforded to you by the DSP Office on campus.

HW3.1 Periodic Periodic Periodic For each signal below, determine if it is periodic or not. If it is, determine the fundamental period. For CT signals, we define $x(t)$, $t \in \mathbb{R}$, and for DT signals, we define $x[n]$, $n \in \mathbb{Z}$.

Recall that the fundamental period T of a continuous signal x is the smallest non-zero real number such that $x(t) = x(t + T)$, $\forall t \in \mathbb{R}$. In the discrete case, it is analogously the smallest positive integer N such that $x[n] = x[n + N]$, $\forall n \in \mathbb{Z}$.

(a) $x(t) = e^{\frac{i\pi t}{2}}$.

$$x(t + T) = e^{\frac{i\pi(t+T)}{2}} = e^{\frac{i\pi t}{2}} \cdot e^{\frac{i\pi T}{2}}.$$

The expression on the right equals $x(t)$ when $e^{\frac{i\pi T}{2}} = 1$, or when $T = 4$.

(b) $x[n] = \cos(n)$.

This is not periodic. For N to be a period of $x[n]$, we must have $x[n + N] = x[n]$, $\forall n \in \mathbb{Z}$:

$$\cos(n + N) = \cos(n) \implies N = 2\pi k, k \in \mathbb{Z}, \forall n \in \mathbb{Z}$$

Because we are constrained to integer periods, there is no integer N such that $x[n + N] = x[n]$. In general, a discrete-time sinusoid is periodic iff the frequency is a rational multiple of π .

(c) $x[n] = \cos(\pi n)$.

For N to be a period of $x[n]$, we must have $x[n + N] = x[n]$, $\forall n \in \mathbb{Z}$:

$$\begin{aligned} \cos(\pi(n + N)) = \cos(\pi n) &\implies \pi N = 2\pi k, k \in \mathbb{Z}, \forall n \in \mathbb{Z} \\ &\implies N = 2k. \end{aligned}$$

The smallest positive integer N for which this holds is $N = 2$. Hence, $\cos(\pi n)$ has a period of 2. This is also equivalent to the signal $(-1)^n$, or the signal $e^{i\pi n}$.

(d) $x(t) = \sin(2\pi t) + \sin(3\pi t)$.

The first sinusoid has a period of 1, and the second a period of $2/3$, so the overall period is the smallest number to be an integer multiple of both 1 and $2/3$, aka 2.

(e) **(Challenge)** $x[n] = \cos\left(\frac{\pi}{2}n^2\right)$.

This is actually periodic! The fundamental period is $N = 2$. Although the continuous-time analog of this function is not periodic (as it oscillates faster and faster towards the edges), the restriction of n to the integers allows $x[n]$ to be periodic.

Consider $x[n + N] = \cos\left(\frac{\pi}{2}(n + N)^2\right) = \cos\left(\frac{\pi}{2}n^2 + \pi Nn + \frac{\pi}{2}N^2\right)$. For N to be a period of $x[n]$, we must have $x[n + N] = x[n]$, $\forall n \in \mathbb{Z}$:

$$\begin{aligned} \cos\left(\frac{\pi}{2}n^2 + \pi Nn + \frac{\pi}{2}N^2\right) = \cos\left(\frac{\pi}{2}n^2\right) &\implies \pi Nn + \frac{\pi}{2}N^2 = 2\pi k, k \in \mathbb{Z}, \forall n \in \mathbb{Z} \\ &\implies Nn/2 + N^2/4 = k. \end{aligned}$$

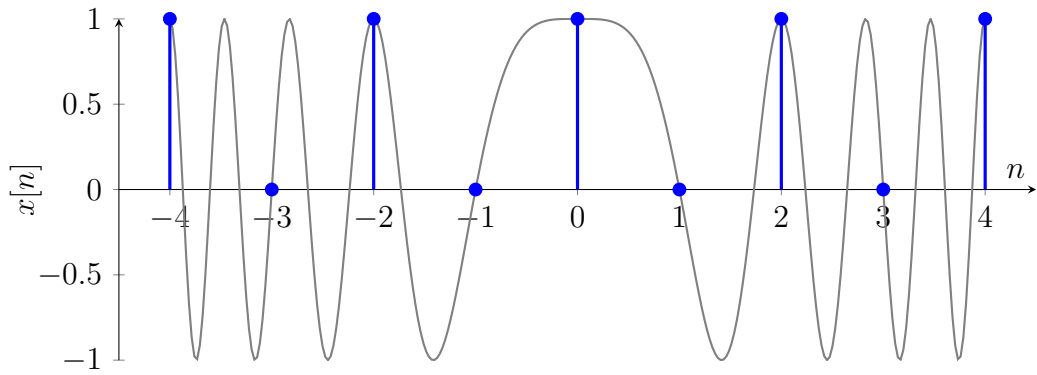
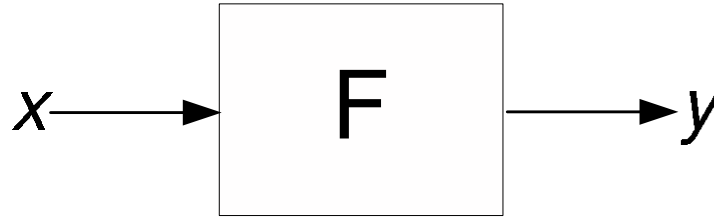


Figure 1: Plot of $x[n]$, with the analogous continuous-time function superposed.

The smallest N for which this holds is $N = 2$. $x[n]$, as well as the continuous analog $x(t)$, is shown in fig:cos-n-sq.

HW3.2 (Frequency Response of a Discrete-Time IIR Filter) The causal, discrete-time LTI filter F shown below



is characterized by the recursive difference equation $y[n] = \alpha y[n - N] + x[n]$, where $|\alpha| < 1$ and $N \in \{1, 2, 3, \dots\}$.

- (a) Determine a simple expression for $f[n]$, where f is the impulse response of the filter. Express your answer in terms of α and N .
 Hint: Because the system is causal, you know $f[n] = 0$ for $n < 0$. You don't need to derive the expression rigorously. Instead, evaluate a few cases for different values of N and look for a pattern.

We know that when we input $\delta[n]$ into our system, it should output the impulse response $f[n]$, giving us the recursive equation:

$$f[n] = \alpha f[n - N] + \delta[n]$$

For $N = 3$,

$$\begin{aligned}
 f[0] &= \alpha f[-3] + \delta[0] = 1 \\
 f[1] &= \alpha f[-2] + \delta[1] = 0 \\
 f[2] &= \alpha f[-1] + \delta[2] = 0 \\
 f[3] &= \alpha f[0] + \delta[3] = \alpha \\
 f[4] &= \alpha f[1] + \delta[4] = 0 \\
 f[5] &= \alpha f[2] + \delta[5] = 0 \\
 f[6] &= \alpha f[3] + \delta[6] = \alpha^2 \\
 f[7] &= \alpha f[4] + \delta[7] = 0 \\
 f[8] &= \alpha f[5] + \delta[8] = 0 \\
 f[9] &= \alpha f[6] + \delta[9] = \alpha^3
 \end{aligned}$$

For $N = 4$,

$$\begin{aligned}
f[0] &= \alpha f[-4] + \delta[0] = 1 \\
f[1] &= \alpha f[-3] + \delta[1] = 0 \\
f[2] &= \alpha f[-2] + \delta[2] = 0 \\
f[3] &= \alpha f[-1] + \delta[3] = 0 \\
f[4] &= \alpha f[0] + \delta[4] = \alpha \\
f[5] &= \alpha f[1] + \delta[5] = 0 \\
f[6] &= \alpha f[2] + \delta[6] = 0 \\
f[7] &= \alpha f[3] + \delta[7] = 0 \\
f[8] &= \alpha f[4] + \delta[8] = \alpha^2 \\
f[9] &= \alpha f[5] + \delta[9] = 0
\end{aligned}$$

Generalizing for N , with $x[n] = \delta[n]$,

$$f[n] = \sum_{k=0}^{\infty} \alpha^k \delta[n - kN]$$

- (b) Determine a reasonably simple expression for $F(\omega)$, where F is the frequency response of the filter. Express your answer in terms of α and N .

We can determine $F(\omega)$ as follows:

$$\begin{aligned}
F(\omega) &= \sum_{n=-\infty}^{\infty} f[n] e^{-i\omega n} \\
&= \sum_{n=-\infty}^{\infty} \sum_{k=0}^{\infty} \alpha^k \delta[n - kN] e^{-i\omega n} \\
&= 1 + \alpha e^{-i\omega N} + \alpha^2 e^{-i\omega 2N} + \alpha^3 e^{-i\omega 3N} + \dots \\
&= \sum_{k=0}^{\infty} (\alpha e^{-i\omega N})^k
\end{aligned}$$

Since $|\alpha| < 1$,

$$F(\omega) = \frac{1}{1 - \alpha e^{-i\omega N}}$$

Alternatively, using the eigenfunction principle, we plug in $x[n] = e^{i\omega n}$ and $y[n] = F(\omega) e^{i\omega n}$ to the LCCDE:

$$y[n] = \alpha y[n - N] + x[n]$$

$$F(\omega)e^{i\omega n} = \alpha F(\omega)e^{i\omega(n-N)} + e^{i\omega n}$$

$$F(\omega)e^{i\omega n} = \alpha F(\omega)e^{-i\omega N}e^{i\omega n} + e^{i\omega n}$$

$$F(\omega)e^{i\omega n} (1 - \alpha e^{-i\omega N}) = e^{i\omega n}$$

$$F(\omega) = \frac{1}{1 - \alpha e^{-i\omega N}}$$

(c) Provide well-labeled sketches of $f[n]$ for

(i) $\alpha = \frac{1}{2}$ and $N = 2$ (ii) $\alpha = \frac{1}{\sqrt{2}}$ and $N = 2$ (iii) $\alpha = \frac{1}{2}$ and $N = 4$.

Describe, in simple terms, the respective effects of α and N on the qualitative features of the impulse response f .

See Figures 2 through 4.

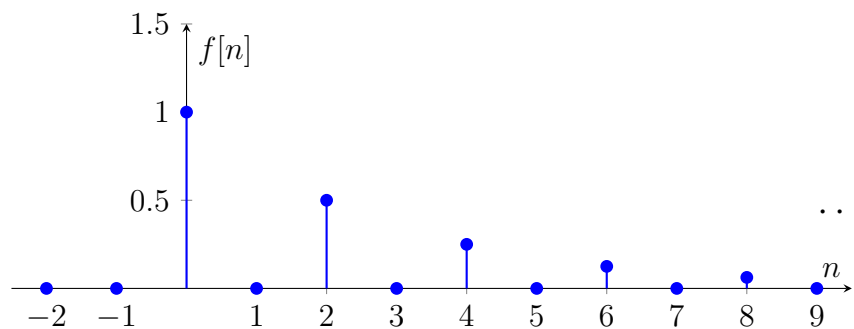


Figure 2: Impulse response for $\alpha = \frac{1}{2}$ and $N = 2$.

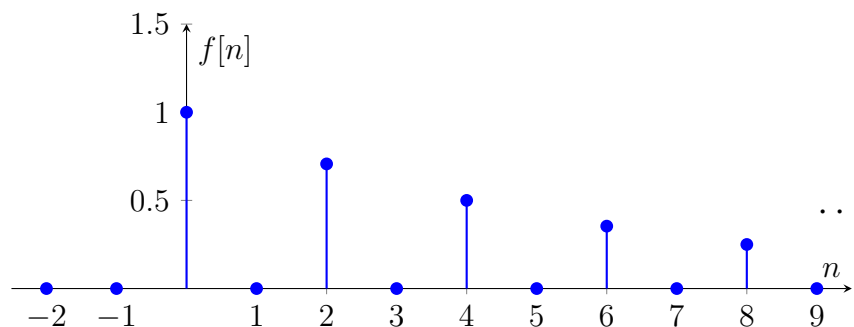


Figure 3: Impulse response for $\alpha = \frac{1}{\sqrt{2}}$ and $N = 2$.

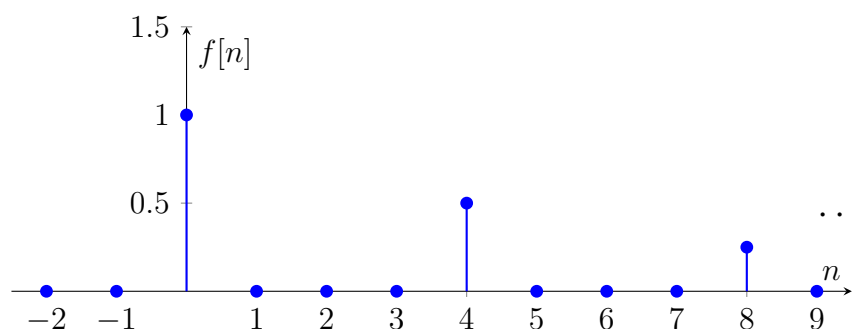


Figure 4: Impulse response for $\alpha = \frac{1}{2}$ and $N = 4$.

From the plots, α is the scale factor that the nonzero values in the impulse response decrease by. N is the number of timesteps between nonzero values in the impulse response, affecting the spacing of nonzero values.

(d) Provide a well-labeled sketch of the magnitude response $|F(\omega)|$ for each of the following cases:

(i) $\alpha = \frac{1}{2}$ and $N = 2$ (ii) $\alpha = \frac{1}{\sqrt{2}}$ and $N = 2$ (iii) $\alpha = \frac{1}{2}$ and $N = 4$.

See Figure 5.

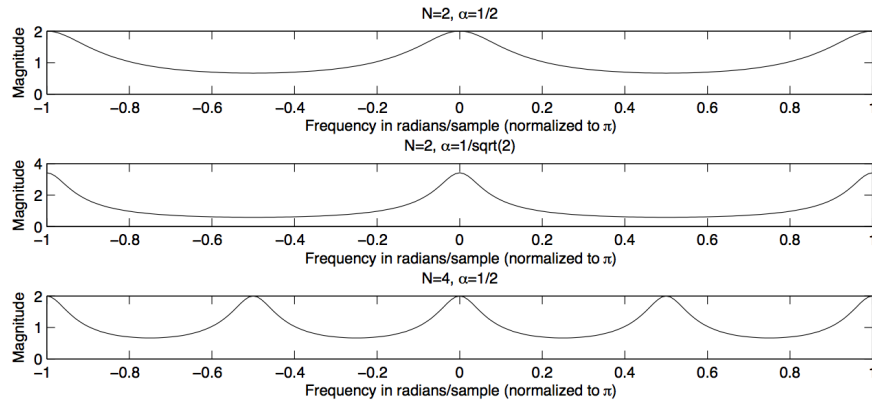


Figure 5: Magnitude responses for 3.1(d).

From the plots, α dictates the range of values the magnitude response can take on, from $\frac{1}{1+\alpha}$ to $\frac{1}{1-\alpha}$. N is how many times the plot repeats.

HW3.3 From LCCDE to Frequency Response and back! This problem will explore the connection between LCCDEs and the frequency response. The LCCDE for system F is:

$$3y[n] - 4y[n-1] - 5y[n-2] = 6x[n] + 7x[n-1] \quad \forall n \in \mathbb{Z}$$

(a) Using the eigenfunction property, find the frequency response $F(\omega)$.

By the eigenfunction property for LTI systems, when we set the input $x[n] = e^{i\omega n}$, then the output $y[n] = F(\omega)e^{i\omega n}$.

$$\begin{aligned} 3y[n] - 4y[n-1] - 5y[n-2] &= 6x[n] + 7x[n-1] \\ 3F(\omega)e^{i\omega n} - 4F(\omega)e^{i\omega(n-1)} - 5F(\omega)e^{i\omega(n-2)} &= 6e^{i\omega n} + 7e^{i\omega(n-1)} \\ F(\omega)(3e^{i\omega n} - 4e^{i\omega(n-1)} - 5e^{i\omega(n-2)}) &= 6e^{i\omega n} + 7e^{i\omega(n-1)} \\ F(\omega)(3e^{i\omega n} - 4e^{i\omega n}e^{-i\omega} - 5e^{i\omega n}e^{-i2\omega}) &= 6e^{i\omega n} + 7e^{i\omega n}e^{-i\omega} \\ F(\omega)(3 - 4e^{-i\omega} - 5e^{-i2\omega}) &= 6 + 7e^{-i\omega} \\ F(\omega) &= \frac{6 + 7e^{-i\omega}}{3 - 4e^{-i\omega} - 5e^{-i2\omega}} \end{aligned}$$

(b) Now given a general LCCDE in this form:

$$a_0y[n] + a_1y[n-1] + \dots + a_ky[n-k] = b_0x[n] + b_1x[n-1] + \dots + b_lx[n-l]$$

Find the frequency response $F(\omega)$.

We can rewrite the above LCCDE above using summations as:

$$\sum_{m=0}^k a_m y[n-m] = \sum_{j=0}^l b_j x[n-j]$$

By the eigenfunction property for LTI systems, when we set the input $x[n] = e^{i\omega n}$, then the output $y[n] = F(\omega)e^{i\omega n}$. Similarly, if have a delayed input $x[n-m] = e^{i\omega(n-m)}$, the delayed output will be $y[n-m] = F(\omega)e^{i\omega(n-m)}$. Plugging this in into (3) gives us:

$$\sum_{m=0}^k a_m F(\omega)e^{i\omega(n-m)} = \sum_{j=0}^l b_j e^{i\omega(n-j)}$$

We can split up each of the complex exponentials, pulling out the $e^{i\omega n}$ part:

$$\sum_{m=0}^k a_m F(\omega)e^{i\omega n}e^{-i\omega m} = \sum_{j=0}^l b_j e^{i\omega n}e^{-i\omega j}$$

We can then pull out the $F(\omega)$ from the left side of the equation, as well as the $e^{i\omega n}$ from both sides to give us:

$$F(\omega)e^{i\omega n} \sum_{m=0}^k a_m e^{-i\omega m} = e^{i\omega n} \sum_{j=0}^l b_j e^{-i\omega j}$$

Finally, cancelling out the $e^{i\omega n}$ from both sides and isolating (ω) gives us a final answer of:

$$\begin{aligned} F(\omega) &= \frac{\sum_{j=0}^l b_j e^{-i\omega j}}{\sum_{m=0}^k a_m e^{-i\omega m}} \\ &= \frac{b_0 + b_1 e^{-i\omega} + \dots + b_l e^{-i\omega l}}{a_0 + a_1 e^{-i\omega} + \dots + a_m e^{-i\omega m}} \end{aligned}$$

- (c) Based on your results from above, find the LCCDE given the frequency response for a new LTI system H.

$$H(\omega) = \frac{3 + 8e^{3i\omega} - e^{i5\omega}}{1 + e^{i\omega}}$$

Before we can pattern match with part (b) to find our solution, we notice that the exponents of the complex exponentials in $H(\omega)$ are positive, and not negative like in part (b). To address this, we divide both the numerator and denominator of $H(\omega)$ by $e^{5i\omega}$:

$$\begin{aligned} H(\omega) &= \frac{\frac{3+8e^{3i\omega}-e^{i5\omega}}{e^{5i\omega}}}{\frac{1+e^{i\omega}}{e^{5i\omega}}} \\ &= \frac{3e^{-5i\omega} + 8e^{-2i\omega} - 1}{e^{-5i\omega} + e^{-4i\omega}} \end{aligned}$$

Now, pattern matching with the equation we found for the frequency response in part (b), we can see that:

$$\begin{aligned} b_5 &= 3 \\ b_2 &= 8 \\ b_0 &= -1 \\ a_5 &= 1 \\ a_4 &= 1 \end{aligned}$$

Plugging these coefficients into (2), we find the following LCCDE:

$$y[n-5] + y[n-4] = 3x[n-5] + 8x[n-2] - x[n]$$

Since an LTI system is uniquely determined from its frequency response, this must be the right one.